

WEEK-2

```
mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE books (
  -> book_id INT AUTO_INCREMENT PRIMARY KEY,
  -> title VARCHAR(255) NOT NULL,
  -> isbn VARCHAR(13) NOT NULL UNIQUE,
  -> published_year INT,
  -> CONSTRAINT chk_published_year CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE members (
  -> member_id INT AUTO_INCREMENT PRIMARY KEY,
  -> full_name VARCHAR(100) NOT NULL,
  -> email VARCHAR(150) NOT NULL UNIQUE
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO books(title, isbn, published_year) VALUES
  -> ('The Alchemist', '9780061122415', 1998),
  -> ('Clean Code', '9780132350884', 2008),
  -> ('Atomic Habits', '9780735211292', 2018);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO members (full_name, email) VALUES
  -> ('Anil Kumar', 'anil.kumar@example.com'),
  -> ('Priya Sharma', 'priya.sharma@example.com'),
  -> ('Ravi Verma', 'ravi.verma@example.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SHOW DATABASES;
```

Database
bookflow_db
information_schema
mysql
performance_schema
studentdb
sys

```
6 rows in set (0.01 sec)
```

```
mysql> SHOW TABLES;
```

Tables_in_bookflow_db
books
members

```
2 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1998
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Members;
```

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com

```
3 rows in set (0.00 sec)
```

```
mysql> USE bookflow_db;
```

```
Database changed
```

```
mysql> CREATE TABLE Loans (
```

```
->     loan_id INT PRIMARY KEY,
```

```
->     member_id INT,
```

```
->     book_id INT,
```

```
->     loan_date DATE,
```

```
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),
```

```
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
```

```
-> );
```

```
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date)
```

```
-> VALUES
```

```
-> (1, 1, 101, '2025-01-05'),
```

```
-> (2, 2, 102, '2025-01-08'),
```

```
-> (3, 3, 103, '2025-01-10'),
```

```
-> (4, 1, 102, '2025-02-01'),
```

```
-> (5, 2, 101, '2025-02-05'),
```

```
-> (6, 3, 102, '2025-02-12'),
```

```
-> (7, 1, 103, '2025-03-01'),
```

```
-> (8, 2, 103, '2025-03-07'),
```

```
-> (9, 3, 101, '2025-03-15'),
```

```
-> (10, 1, 101, '2025-04-01');
```

```
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint  
' REFERENCES `books` (`book_id`))
```

```
mysql> SELECT * FROM Books;
```

```
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title       | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist | 9780061122415 | 1998           |
| 2       | Clean Code   | 9780132350884 | 2008           |
| 3       | Atomic Habits | 9780735211292 | 2018           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name   | email                     |
+-----+-----+-----+
| 1         | Anil Kumar  | anil.kumar@example.com   |
| 2         | Priya Sharma | priya.sharma@example.com |
| 3         | Ravi Verma  | ravi.verma@example.com   |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> USE bookflow_db;
Database changed
mysql> DROP TABLE IF EXISTS Donation_History;
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql> DROP TABLE IF EXISTS Loans;
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Loans (
->     loan_id INT PRIMARY KEY,
->     member_id INT,
->     book_id INT,
->     loan_date DATE,
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date)
-> VALUES
-> (1, 1, 1, '2025-01-05'),
-> (2, 2, 2, '2025-01-08'),
-> (3, 3, 3, '2025-01-10'),
-> (4, 1, 2, '2025-02-01'),
-> (5, 2, 1, '2025-02-05'),
-> (6, 3, 2, '2025-02-12'),
-> (7, 1, 3, '2025-03-01'),
-> (8, 2, 3, '2025-03-07'),
-> (9, 3, 1, '2025-03-15'),
-> (10, 1, 1, '2025-04-01');
```

```
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	1	1	2025-01-05
2	2	2	2025-01-08
3	3	3	2025-01-10
4	1	2	2025-02-01
5	2	1	2025-02-05
6	3	2	2025-02-12
7	1	3	2025-03-01
8	2	3	2025-03-07
9	3	1	2025-03-15
10	1	1	2025-04-01

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT
->     m.full_name AS Member_Name,
->     b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m
-> ON l.member_id = m.member_id
-> INNER JOIN Books b
-> ON l.book_id = b.book_id;
```

Member_Name	Book_Title
Anil Kumar	The Alchemist
Anil Kumar	Clean Code
Anil Kumar	Atomic Habits
Anil Kumar	The Alchemist
Priya Sharma	Clean Code
Priya Sharma	The Alchemist
Priya Sharma	Atomic Habits
Ravi Verma	Atomic Habits
Ravi Verma	Clean Code
Ravi Verma	The Alchemist

10 rows in set (0.00 sec)

```
mysql> SELECT
->     published_year,
->     COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1998	1
2008	1
2018	1

3 rows in set (0.00 sec)

```
mysql> CREATE TABLE Donation_History (
->     donation_id INT PRIMARY KEY,
->     book_id INT,
->     donor_name VARCHAR(100),
->     donation_date DATE,
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> START TRANSACTION;
```

Query OK, 0 rows affected (0.00 sec)

```
mysql>
```

```
mysql> INSERT INTO Books
```

```
-> (book_id, title, isbn, published_year)
```

```
-> VALUES
```

```
-> (4, 'Animal Farm', '9780451526342', 1945);
```

Query OK, 1 row affected (0.00 sec)

```
mysql>
```

```
mysql> INSERT INTO Donation_History
```

```
-> (donation_id, book_id, donor_name, donation_date)
```

```
-> VALUES
```

```
-> (1, 4, 'Raj Kumar', CURDATE());
```

Query OK, 1 row affected (0.00 sec)

```
mysql>
```

```
mysql> COMMIT;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1998
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018
4	Animal Farm	9780451526342	1945

4 rows in set (0.00 sec)

```
mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
| 1 | 4 | Raj Kumar | 2026-07-30 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 1 | Anil Kumar | anil.kumar@example.com |
| 2 | Priya Sharma | priya.sharma@example.com |
| 3 | Ravi Verma | ravi.verma@example.com |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> v
->
-> SELECT * FROM Loans;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual th
```

```
SELECT * FROM Loans' at line 1
mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 1 | 1 | 2025-01-05 |
| 2 | 2 | 2 | 2025-01-08 |
| 3 | 3 | 3 | 2025-01-10 |
| 4 | 1 | 2 | 2025-02-01 |
| 5 | 2 | 1 | 2025-02-05 |
| 6 | 3 | 2 | 2025-02-12 |
| 7 | 1 | 3 | 2025-03-01 |
| 8 | 2 | 3 | 2025-03-07 |
| 9 | 3 | 1 | 2025-03-15 |
| 10 | 1 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Donation_History;
```

donation_id	book_id	donor_name	donation_date
1	4	Raj Kumar	2026-07-30

```
1 row in set (0.00 sec)
```